Supplementary Materials

Scaled CO Electroreduction to Alcohols

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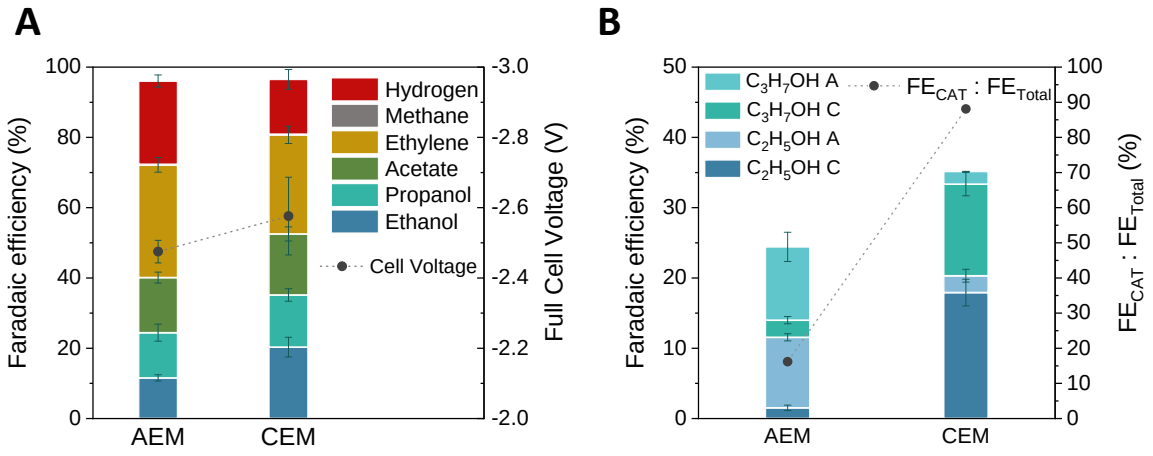
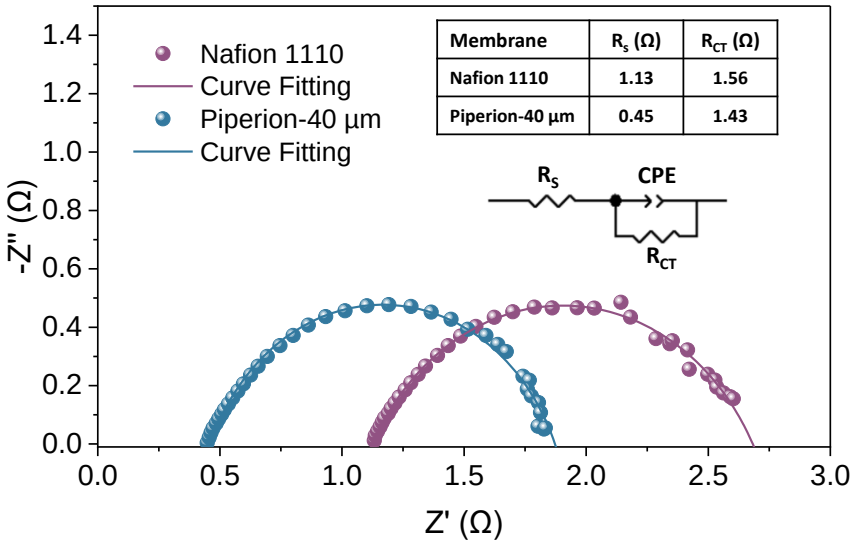
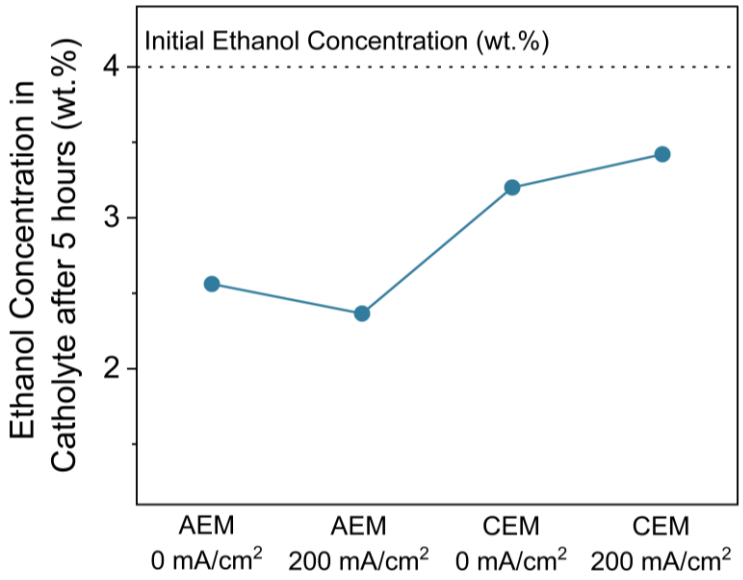


Supplementary Figure 1. (A) Total product distributions for the baseline AEM (PiperION-40 µm) and CEM (Nafion 1110) tests. **(B)** Alcohol faradaic efficiencies collected from the anode and the cathode sides respectively for the baseline AEM and CEM COR tests. The tests were operated at 200 mA/cm^2. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates). Voltages represent the full-cell MEA potential and are not iR-corrected. Source data for the results are provided as a Source Data file.



Supplementary Figure 2. Electrochemical impedance spectroscopes (EIS) measurements for the CEM (Nafion 1110) and AEM (PiperION-40 µm) MEA cells. EIS was carried out under the same conditions as the 5 cm^2 tests, with CO gas purging during the measurement.



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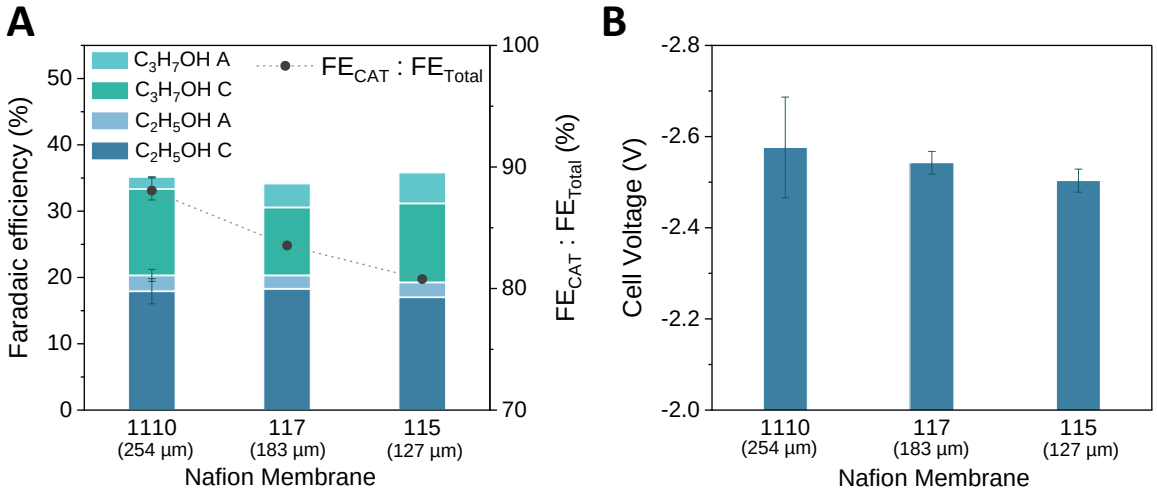

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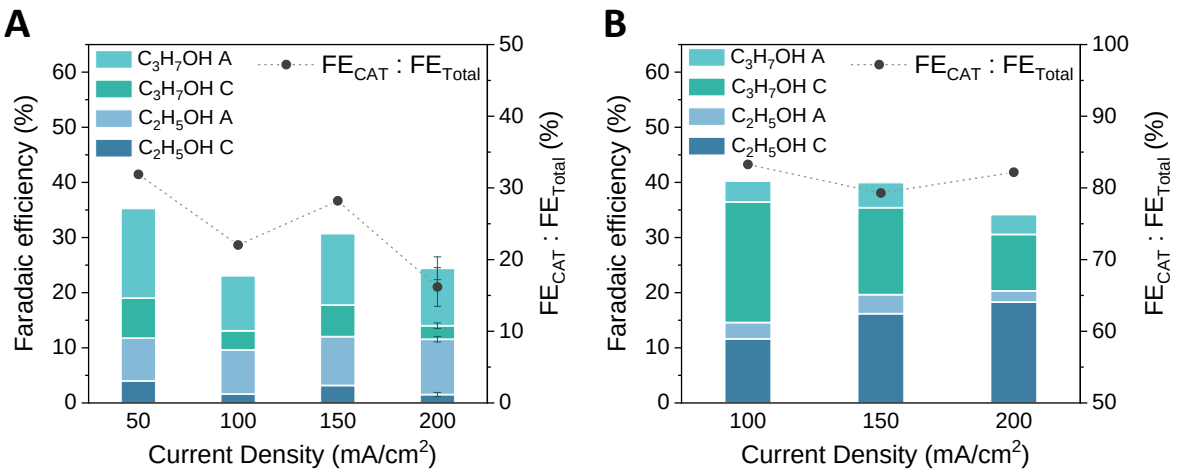
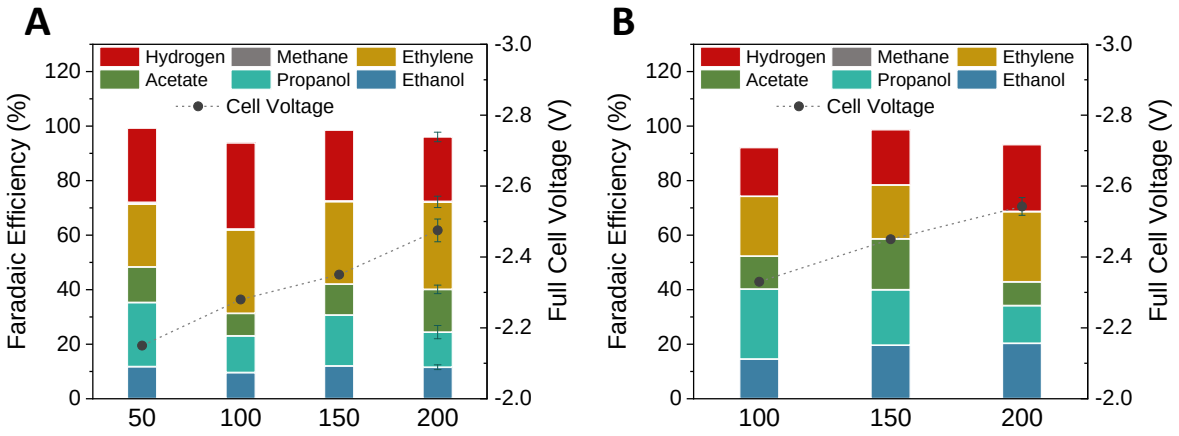


Supplementary Figure 5. Alcohol faradaic efficiencies collected from the anode and the cathode sides respectively at different current densities of the (A) AEM MEA (PiperION-40 µm) and **(B)** CEM MEA (Nafion 117). Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates). Source data for the results are provided as a Source Data file.



Supplementary Figure 6. Product faradaic efficiencies of the (A) AEM MEA (PiperION-40 µm) and **(B)** CEM MEA (Nafion 117). Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates). Voltages represent the full-cell MEA potential and are not iR-corrected. Source data for the results are provided as a Source Data file.

Supplementary Note 1. TEA Analysis on different CO_2 to ethanol electrolyzer systems.

A TEA analysis was conducted to compare the economic viability of generating ethanol from CO_2 electrolysis using different electrolyzer configurations. The TEA calculation is based on our recent publication on cascade CO_2/CO electrolysis[1]. However, the calculations have been modified to consider the production of 1 tonne of ethanol per day instead of ethylene at ideal electrolyzer conditions.

The TEA analysis considers three scenarios for ethanol production: (1) A neutral MEA electrolyzer; (2) SOEC to an alkaline AEM MEA electrolyzer; (3) SOEC to an alkaline CEM MEA electrolyzer. (system used in this work). In order to keep a fair comparison for the future industrial application of CO_2 electroconversion, we considered the electrolyzer voltage and faradaic efficiencies at ideal conditions. The ethanol concentrations and crossover fractions were obtained experimentally (**Supplementary Tables 1 and Supplementary Figure 7**). Since the cathode-outlet concentrations were similar, the values were averaged to ~4 wt% in order to simplify the ethanol distillation model. The crossed-over ethanol in the anode stream was assumed to have a concentration of ~0.05 wt%, keeping consistent with previous work[2]. The multi-step condenser strategy was not considered in this analysis. The other input parameters, including those for the SOEC, were obtained from our previous publication[1]. All input parameters can be found in **Supplementary Tables 2 and 3**. The ethanol distillation calculations are obtained from an OLI separation model, the details of which are provided in the next section. Please refer to our previous publication for all other calculation costs.[1] The final costs are presented in **Supplementary Table 4.**

Supplementary Table 1. Ethanol concentrations of the cathode stream for the AEM CO_2, AEM CO, and CEM CO electrolyzer tests.

Parameter	CO_2 AEM	AEM CO	CEM CO
Cathode-side Ethanol Conc. (wt%)	3.8	4.05	4.3

Supplementary Table 2. Input electrolyzer parameters for the TEA analysis. Values for the AEM and CEM MEAs are obtained experimentally.

Parameter	CO_2 AEM	SOEC[1]	AEM CO	CEM CO
Cell Voltage (V)	3	1.4	2	2.1
FE (%) (CO or EtOH)	95	95	95	95
Current Density (mA/cm^2)	200	800	200	200
CO_2 Utilization (%)	25[3]	90	90	90
Cathode-side Ethanol Conc. (wt%)	4	N/A	4	4
Anode-side Ethanol Conc. (wt%)	0.05	N/A	0.05	0.05
Ethanol Crossover (%)	78	N/A	84	12

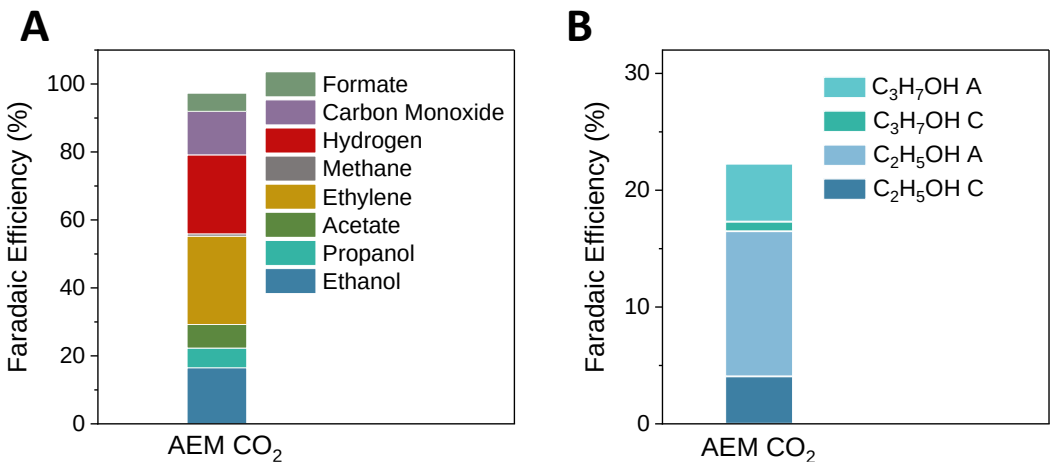
Supplementary Table 3. Input economic parameters for the TEA analysis. Vales obtained from previous publication[1], unless otherwise indicated.

Parameter	Value
CO_2 Capture ($/t$CO_2$)	30
System Lifetime (years)	30
Catalyst/membrane lifetime (years)	5
Electrolyte lifetime (years)	1
Balance of Plant (%)	50
Lang factor	1
Capacity factor	0.9
CO_2/CO MEA cost ($/kW)	300

SOEC cost ($/kW)	250
Energy cost ($/kWh)	0.02[4]
H_2O Cost ($/t)	5
$KHCO_3$/KOH Cost	750/1000

Supplementary Table 4. Summary of all costs based on the input parameters above on a $/tEtOH basis.

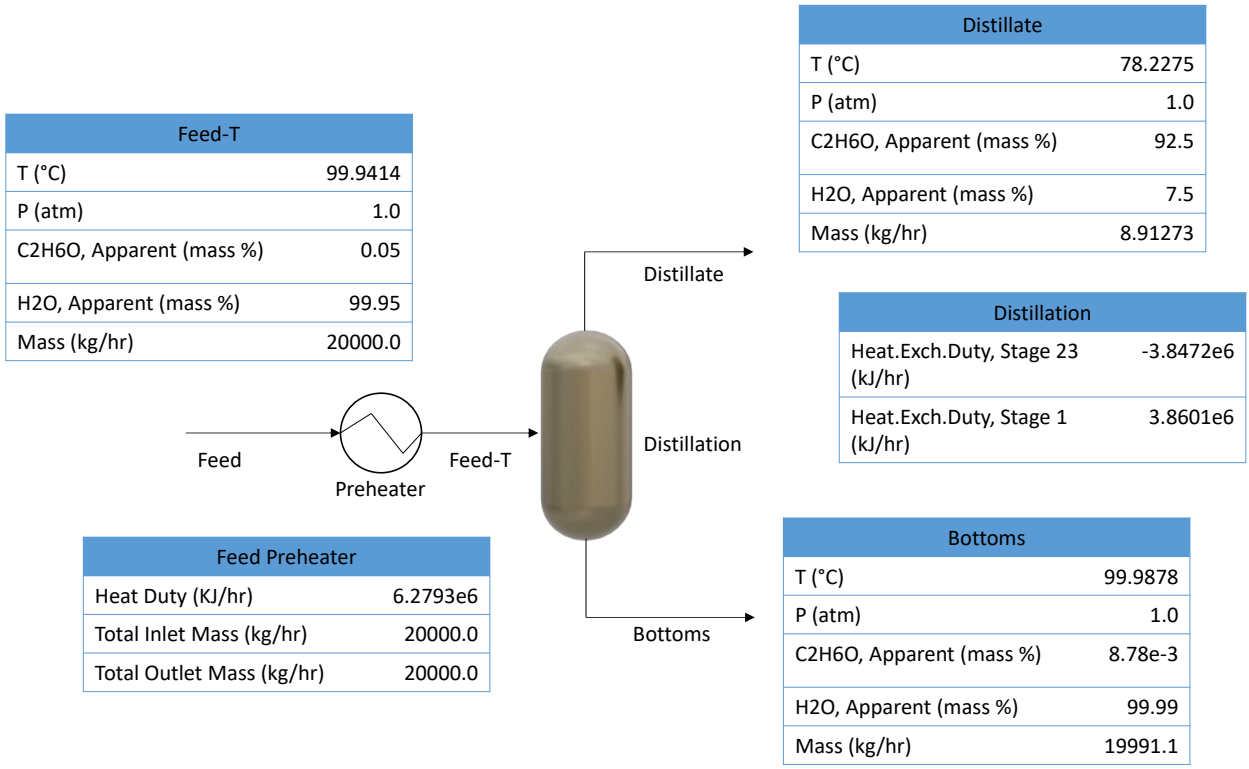
Parameter	**CO_2 AEM**	**CO AEM**	**CO CEM**
Anode Distillation	2103	2264	323
Cathode Distillation	8	6	33
EC Operation	498	227	238
CO_2 Capture	229	64	64
Capital Costs & Installation	354	171	171
SOEC Cell	0	143	143
Total $/t EtOH	3192	2875	972



Supplementary Figure 7. (A) Total product distributions for the AEM CO_2R test. The average cell voltage was approximately 3.5 V. **(B)** Alcohol faradaic efficiencies collected from the anode and the cathode sides respectively for the AEM CO2R test. The electrolyzer was operated under the same conditions as the COR tests at a current density of 200 mA/cm^2 with PiperION-40 µm. The cell retained approximately 22% of alcohols on the cathode side, meaning 78% of alcohols crossed over the AEM membrane to the anolyte. Source data for the results are provided as a Source Data file.

Ethanol Separation Model:

The ethanol separation step was modeled as a conventional distillation process in OLI Flowsheet V11.5 in order to estimate the energy required to separate an ethanol-water feed at varying ethanol concentrations. This distillation process was modeled under the following assumptions: a simple feed stream containing only water and ethanol; ideal equilibrium column stages; a constant column pressure of 1 atm; and a constant total ethanol flowrate of 10 kg/h in the feed stream across all ethanol feed concentrations. In addition, the feed stream entered the process at 25 °C and was preheated to a saturated liquid state before entering the column. The separation of ethanol and water was specified based on mass fraction compositions. The target distillate ethanol concentration was 92.5 wt.% and the target bottoms ethanol concentration was 0.008 wt.%. The OLI separation model is shown in **Supplementary Figure 8**. `



Supplementary Figure 8. OLI distillation model for the ethanol-water separation cost calculation.

The following approach was used to design the distillation column:

1. Set the total number of stages (20, 30, 40, and 50 stages were investigated for each feed).
2. Set a feed stage.
3. Adjust the reboiler and condenser duties until the desired separation is achieved.
4. Plot a McCabe-Thiele diagram using the calculated column data.
5. Note any pinched stages and compare the current feed stage with the stage that intersects with the feed line. Since the feed is a saturated liquid, the feed line will be vertical perpendicular to the x-axis.
6. If pinched stages are present and/or the current feed stage is not the same as the stage that intersects with the feed line, shift the feed stage towards the pinched stages and towards the feed line; this will usually result in a shift in the same direction.
7. Repeat steps 3 to 6 until the optimal feed location is achieved. This will coincide with the lowest heat duties and lowest reflux ratio required for the current total number of stages.

Once an optimal case had been found for each total number of stages for a given feed, we assumed that the 50-stage column was operating under the minimum reflux ratio (R_{min}). In order to select a final design, an optimal reflux ratio of $1.2*R_{min}$ was chosen. This was done to provide a margin of safety by operating above the minimum reflux ratio while also limiting the number of stages. The calculated reflux ratios for each feed and total number of stages are shown in **Supplementary Table 5.**

Supplementary Table 5. The reflux ratio for each feed concentration and number of stages.

Ethanol Feed Concentration (wt%)	20 Stages	30 Stages	40 Stages	50 Stages	$1.2*R_{min}$
10	3.19	2.25	2.18	2.18	2.62
4	7.31	5.59	5.16	5.04	6.048
0.05	404.60	386.15	383.53	383.06	459.67

The simulation results for the final designs are reported below in **Supplementary Tables 6 and 7.** The energy costs are normalized per kg of ethanol in the distillate stream. The total number of stages for each scenario was selected to be 23, and a feed stage that provided a reflux ratio close to the calculated optimal was selected. Since the total number of stages is discrete, the optimal reflux ratio could not be achieved precisely.

Supplementary Table 6. Final distillation column results for each feed case.

Ethanol Feed Concentration (wt%)	Feed Stage	Total Stages	Reflux Ratio (mol/mol)	Preheater Duty (kJ/kg EtOH)	Reboiler Duty (kJ/kg EtOH)	Condenser Duty (kJ/kg EtOH)
10	12	23	2.71	2785.83	4586.44	-3158.45
4	14	23	6.32	7441.92	8866.97	-7365.64
0.05	13	23	394.30	749544.17	460768.54	-459206.01

Supplementary Table 7. Mass flowrates and overall ethanol recovery in the distillate stream for each feed case.

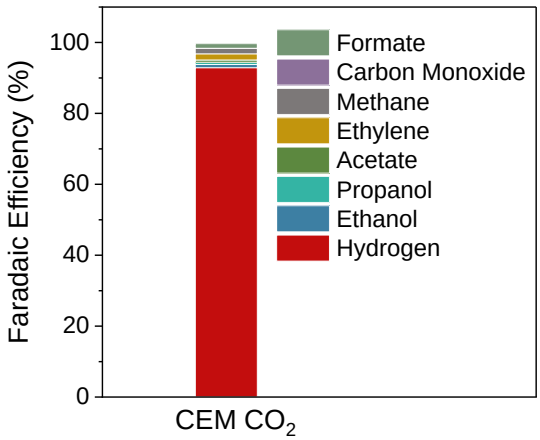
Ethanol Feed Concentration (wt%)	Feed Flow (kg/h)	Distillate Flow (kg/h)	Bottoms Flow (kg/h)	Ethanol Recovery (%)
10	100	10.8029	89.1968	99.93
4	250	10.7903	239.209	99.81
0.05	20000	9.05669	19990.9	83.77

There are several disadvantages when utilizing lower ethanol feed concentrations. First, the separation requires a higher reflux ratio, which increases the energy cost. Furthermore, a greater total feed flowrate adds to the required preheater and reboiler loads. Lastly, the larger amount of water present in the system produces a larger bottoms flowrate, and therefore a greater loss of ethanol for the same bottoms ethanol concentration (**Supplementary Table 7**.).

The distillation operating costs were obtained from the required low-pressure steam (LPS) and cooling water flow rates. A heat capacity of 4.186 kJ/kg°C and enthalpy of vaporization of 2021.09 kJ/kg were used for cooling water and LPS[5]. The cost of cooling water was assumed to be $0.04/tonne[6]. The cost of LPS was assumed to be $4.7/tonne[7]. Values are converted to 2024 USD. The final utility costs are presented in **Supplementary Table 8**.

Supplementary Table 8. Utility costs for the ethanol separation model.

Ethanol Feed Concentration (wt%)	Cooling Water (t/tEtOH)	Low-Pressure Steam (t/tEtOH)	Low-Pressure Steam Cost ($/tEtOH)	CW Cost ($/tEtOH)	Total Utilities Cost ($/tEtOH)
10	83.84	3.65	13.48	3.32	16.80
4	195.51	8.07	29.82	7.74	37.56
0.05	12188.94	598.84	2212.95	482.68	2695.64



Supplementary Figure 9. Total product distributions for the CEM CO_2R test. The average cell voltage was approximately 3.5 V. The electrolyzer was operated under the same conditions as the COR tests at a current density of 200 mA/cm^2. Source data for the results are provided as a Source Data file.

Supplementary Note 2. Comparison of different works investigating CO_2/CO to ethanol electrolysis considering production rate and cathodic ethanol fraction.

The data used for the construction of Figure 1D in the main text is presented in **Supplementary Table 9.** The production rates were calculated using the reported cell sizes, current densities, and FEs from the reported works. The cathodic ethanol fractions for both alkaline flow cells and AEM MEAs are not typically reported. Instead, we average the results reported by Seger et al.[8] (~6%) and Züttel et al.[9] (~30%) to obtain an approximate cathodic ethanol fraction for alkaline flow cells of 18%. A cathode ethanol fraction of approximately 30% was previously reported by Gabardo et al. for the AEM MEA[10].

Supplementary Table 9. The production rate and ethanol concentration of various CO_2 and CO electroreduction systems.

Reference	Electrolyzer System	Cell Size	Production Rate (mL/day)	Cathodic Ethanol Fraction (%)
This work	**CEM MEA**	**800 cm^2**	**96.17**	**65**
This work	**CEM MEA**	**50 cm^2**	**9.68**	**88**
Joule (2021) [2]	SSE MEA	5 cm^2	0.45	60
Joule (2020) [10]	Neutral MEA	5 cm^2	0.52	30
Nat. Energy (2022) [11]	Neutral MEA	5 cm^2	1.84	30
ACS EL. (2022) [12]	Neutral MEA	5 cm^2	1.09	30
Nat. Energy (2022) [11]	Alkaline Flow Cell	1 cm^2	0.68	18
Nat. Comm. (2023) [13]	Alkaline Flow Cell	1 cm^2	0.48	18
App. Cat. B. (2023) [14]	Alkaline Flow Cell	1 cm^2	0.40	18
Nat. Catal. (2022) [15]	Alkaline Flow Cell	1 cm^2	0.83	18
Angew (2022) [16]	Alkaline Flow Cell	1 cm^2	0.89	18

Supplementary Table 10. Input values for the energy analysis of different ethanol producing MEAs from CO_2/CO. Since not all values were included in each study, some assumed values were chosen based on other publications and referenced in the table.

Parameter	**CO_2 SSE[2]**	**CO_2 AEM[10]**	**CO_2 AEM[17]**	**SOEC[1]**	**CO AEM**
Cell Voltage (V)	3.8	3.9	3.5	1.4	2.58
FE (%) (CO or EtOH)	20%	23%	50%	95%	20%
Current Density (mA/cm^2)	200	300	200	800	200
CO_2 Utilization (%)	50%	15%	15%[10]	90	18%[1]
Cathode-side Ethanol Conc. (wt%)	13%	4.00%	4.00%	n/a	4.00%
Anode-side Ethanol Conc. (wt%)	0.05%	0.05%	0.05%[10]	n/a	0.05%
Ethanol Crossover (%)	40%	70%	70%[10]	n/a	12%

Supplementary Table 11. Energy analysis of ethanol producing MEAs from CO_2/CO. The anode and cathode ethanol distillation energy requirements were calculated in **Supplementary Table 6.** Minimizing the amount of anode distillation energy needed is critical to the overall energy requirement of the electrolyzer. The two-stage condenser is not needed for the SSE system, as the concentrated ethanol is recovered in a separate midchannel instead of the cathode gas outlet. The energy cost for the two-stage condenser is shown in **Supplementary Note 5.**

Parameter	**CO_2 SSE**	**CO_2 AEM**	**CO_2 AEM**	**CO AEM (This work)**
Anode distillation	300	526	526	90
Cathode distillation	5	3	3	8
EC operation	478	426	176	216
CO_2 capture	8	8	8	9
Cathode gas separation	8	20	20	5
Anode gas separation	0	23	23	0
SOEC cell	0	0	0	23
Two-Stage condenser	0	5	5	14
Total GJ/tEtOH	**799**	**1010**	**760**	**364**

Supplementary Note 3. Cathode-side ethanol separation using a cold-trap.

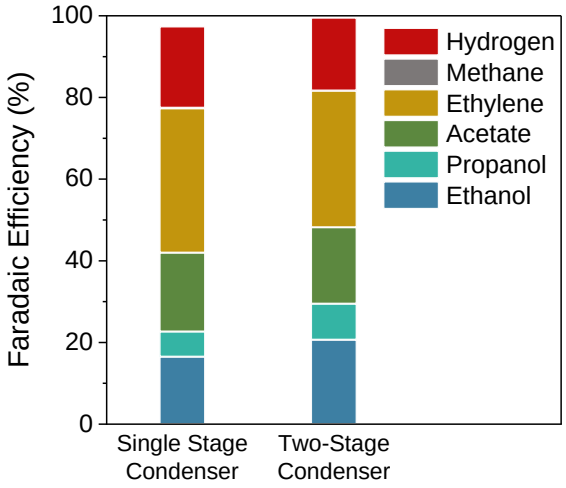
The conventional method to measure the cathode-side alcohols in a CO_2R/COR electrolyzer is to use a cold-trap at the cathode gas outlet[2,10,12]. The gas outlet is bubbled through ice-water to separate the gas-phase alcohols by taking advantage of the greater solubility of alcohols in water compared to the other gas components in the stream (CO, H_2, and C_2H_4). Consequently, the alcohol products are diluted in the final solution due to the addition of water during the bubbling step. To determine the alcohol concentrations before the dilution step, a theoretical concentration is back-calculated based on the amounts of water and ethanol in the cathode gas outlet prior to the liquid-collector. These values are obtained by observing the change in volume of the liquid-collector and the alcohol concentrations after the collection procedure. As an example, we show the calculations for both the measured and theoretical ethanol concentrations for one of our tests using the 5 cm^2 CEM CO electrolyzer.

According to the NMR analysis, the ethanol concentration in the liquid collector after the test was ~0.404 wt%. The theoretical ethanol concentration in prior to dilution in the liquid collector (LC) was then calculated as follows:

$$EtOH\ wt\% = \frac{EtOH\ concentration\ in\ LC \times Final\ Volume\ of\ LC\ \times MW_{EtOH}}{(LC\ Final\ Volume - LC\ Initital\ Volume)\ \times\ \rho_{H_2O}} \quad (1)$$

$$EtOH\ wt\% = \frac{0.109\ \left[\frac{mol}{L}\right] \times \frac{1}{1000}\left[\frac{L}{mL}\right] \times 11.1\ [mL] \times 46.07\ [\frac{g}{mol}]}{(11.1\ [mL] - 10\ [mL]) \times 0.997\ \ [\frac{g}{mL}]} = 4.09\ wt\%$$

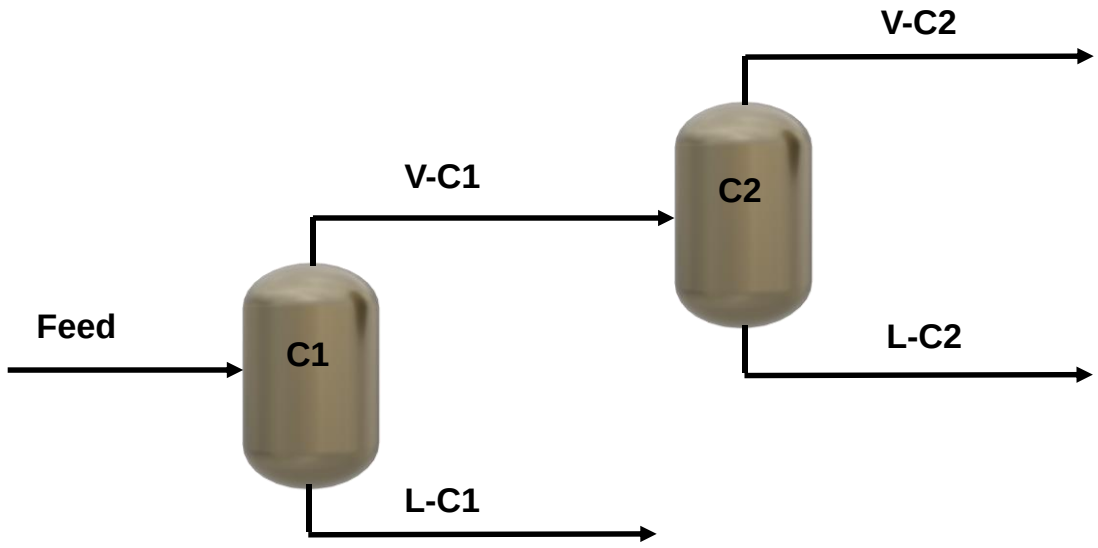
Therefore, the actual ethanol concentration in collector is an order of magnitude lower than the theoretical concentration.



Supplementary Figure 10. Total product distributions for the single and two-stage condenser strategies applied to the 50 cm^2 cell. The missing FE corresponds to the alcohols not fully condensed in the first stage. The cells were operated at 200 mA/cm^2 with Nafion 1110 membrane. Source data for the results are provided as a Source Data file.

Supplementary Note 4. Aspen Plus simulations of the ethanol condensers.

In order to investigate the usage of condensers to separate ethanol and water from the cathode gas stream, we modeled the process in AspenPlus V12.1 (AspenTech Inc., Bedford, MA). The non-random two-liquid model (NRTL) model was chosen because the liquid phase primarily consists of only water and ethanol. The model parameters were obtained from the Aspen database. The Flash2 unit block was used to model the two condensers. The inlet feeds to the simulated condensers were chosen based on the compositions of the cathode gas outlet for the 500 sccm test for the 50 cm^2 cell (**Supplementary Table 12**). The inlet feed was assumed to be 1000 kg/hr at 25°C and 1 bar. Methane was not considered because its faradaic efficiency was negligible in all cases (< 0.5%). The first condenser was maintained at 10 °C. The second condenser temperature was varied between 0 – -40 °C. The separation model is shown in **Supplementary Figure 11**.



Supplementary Figure 11. Representation of the Aspen Plus model for the ethanol and water separation calculations from the gas feed. V-C1 and V-C2 represent the vapour flows from the first and second condensers. L-C1 and L-C2 represent the liquid flows from the first and second condensers.

Supplementary Table 12. Total product distribution at the cathode gas outlet for the different flow rates applied to the 50 cm^2 cell. Values obtained experimentally.

Component	Mass Fraction in Gas Stream
Water	19.14 wt%
Ethanol	0.96 wt%
H_2	0.09 wt%
C_2H_4	1.07 wt%
CO	78.74 wt%

Single-Condenser Scenario:

We consider the 50 cm^2 cell test (**Figure 3A**) with a single condenser at 10 °C. The results from the condenser simulation are shown in **Supplementary Table 13**. Approximately ~30% of the ethanol does not condense in the condenser, instead it remains in the gas phase due to the high CO flow rate acting as the carrier gas. Therefore, a second low-temperature condenser is needed to capture the remaining ethanol (F**igure 3C**).

Supplementary Table 13. Aspen simulation results from the single condenser scenario.

Component	Feed (kg/hr)	V-Cl (Vapour) (kg/hr)	L-C1 (Liquid) (kg/hr)
Water	191.4	6.3	185.1
Ethanol	9.6	2.5	7.1

H2	0.9	0.9	0.0
C2H4	10.7	10.7	0.0
CO	787.4	787.4	0.0

Supplementary Note 5. Energy analysis of the multi-step condenser strategy.

The energy required for the condenser operation was obtained directly from the Aspen Plus Energy Analyzer. The 10 °C and -78 °C condensers require approximately 5.45 GJ/tEtOH and 9.92 GJ/tEtOH of energy input respectively. The energy input for the CEM MEA, assuming a cell voltage of 2.5 V (V) and ethanol faradaic efficiency (FE) of 20%, is approximately 209 GJ/tEtOH. Furthermore, the second condenser temperature implemented in this study is significantly colder than needed to achieve the desired ethanol separation due to the cooling methodology used in the lab. A higher temperature of -30°C (**figure 3C**) would still condense > 95% of ethanol in the cathode gas outlet and require a lower energy input of approximately 6.26 GJ/tEtOH energy input into the second condenser. Consequently, the energy required to operate the second condenser is approximately 3% of the total energy input. The calculation details are shown below.

Calculation Details

To calculate the energy input to the condensers per tonne of ethanol, the net heat duty to each condenser was compared to the total amount of ethanol condensed from the gas stream. The net heat duty and outlet ethanol flowrate captured in the liquid phase are shown in **Supplementary Table 14** for each condenser**.**

Supplementary Table 14. Net heat duty and ethanol flowrate from the liquid phase for each condenser. Data obtained directly from Aspen Plus.

Condenser	Net Heat Duty (GJ/h)	Ethanol in liquid phase (kg/h)
Condenser 1	-0.052	7.1
Condenser 2	-0.095	2.5

The energy input to each condenser is calculated as follows:

$$Energy\ input\ to\ condenser = \frac{Heat\ duty\ \left[\frac{GJ}{h} EtOH\right] \times 1000\ [\frac{kg}{h}]}{Total\ ethanol\ condensed\ \left[\frac{kg}{h}\right]} \quad (2)$$

The values for the second condenser are shown:

$$Energy\ input\ to\ condenser = \frac{-0.095\ \left[\frac{GJ}{h} EtOH\right] \times 1000\ \left[\frac{kg}{h}\right]}{7.1 + 2.5\ \left[\frac{kg}{h}\right]} = 9.92 \left[\frac{GJ}{tEtOH}\right]$$

To calculate the energy input to the electrolyzer, the total charge input per metric tonne of ethanol is required:

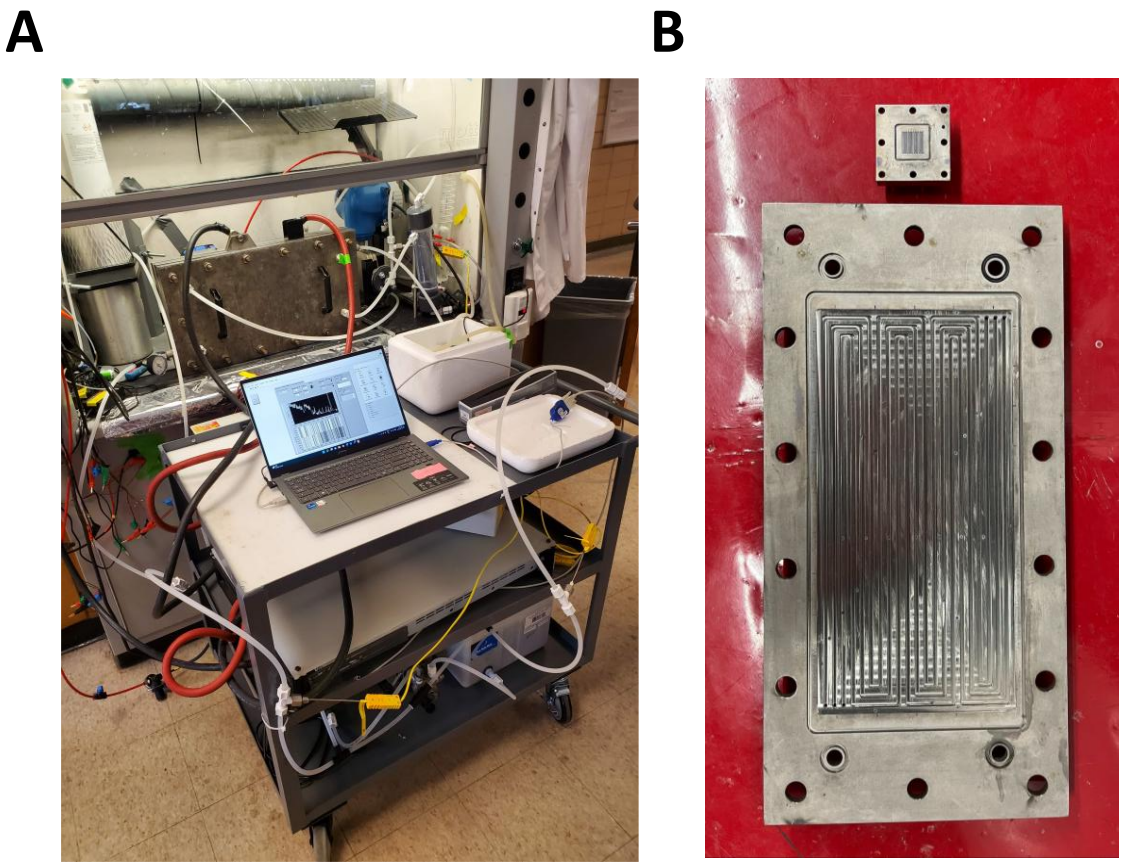
$$Charge\ per\ tonne\ EtOH = \frac{8\ \left[\frac{mol\ e^-}{mol\ EtOH}\right] \times 96485\ [\frac{C}{mol\ e^-}]}{FE_{EtOH}\ \times\ MW_{EtOH}\ \times 10^3} \quad (3)$$

$$Charge\ per\ tonne\ EtOH = \frac{8\ \left[\frac{mol\ e^-}{mol\ EtOH}\right] \times 96485\ \frac{C}{mol\ e^-}}{0.2\ \times\ 46.068\ \left[\frac{g}{mol}\right] \times 10^3} = 83.78\ [\frac{GC}{tEtOH}]$$

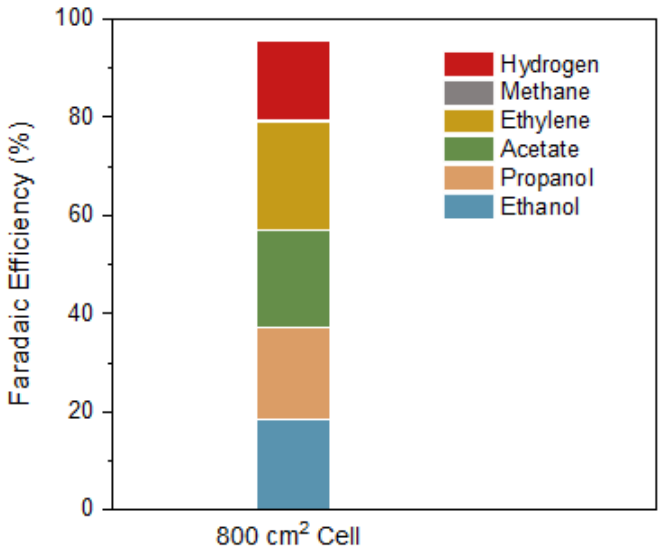
The energy input is calculated by multiplying the charge needed by the cell voltage:

$$Energy\ Input\ per\ tonne\ EtOH = Charge\ per\ tonne\ EtOH \times V = 83.78 \left[\frac{GC}{tEtOH}\right] \times 2.5\ V$$

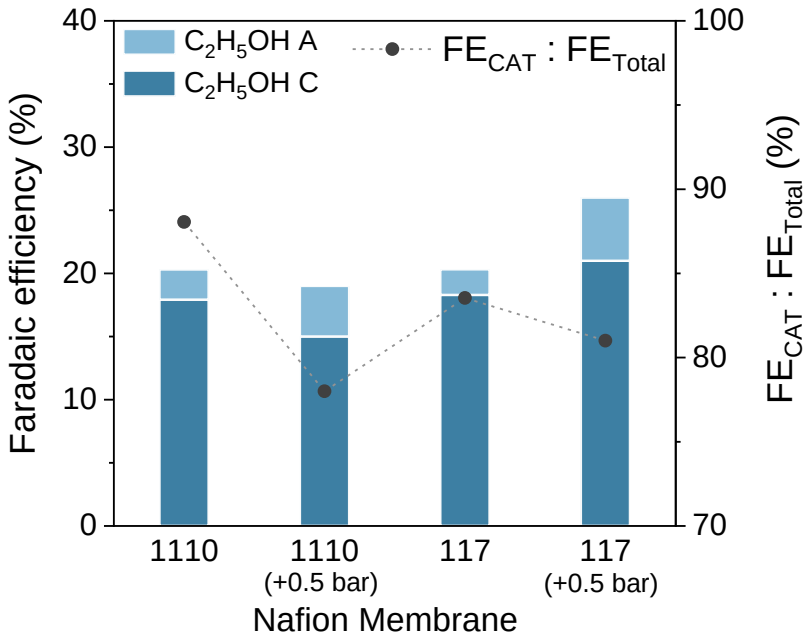
$$= 209.44 \left[\frac{GJ}{tEtOH}\right]$$



Supplementary Figure 12. (A) Photo of the 800 cm^2 cell CO electrolyzer running in the lab fume hood. The cell was run from a control-system placed directly in-front of the fume hood. The three condensers were placed to the right and behind the cell to collect liquid products. **(B)** Size comparison of the 5 cm^2 and 800 cm^2 cells used for the reported COR tests.



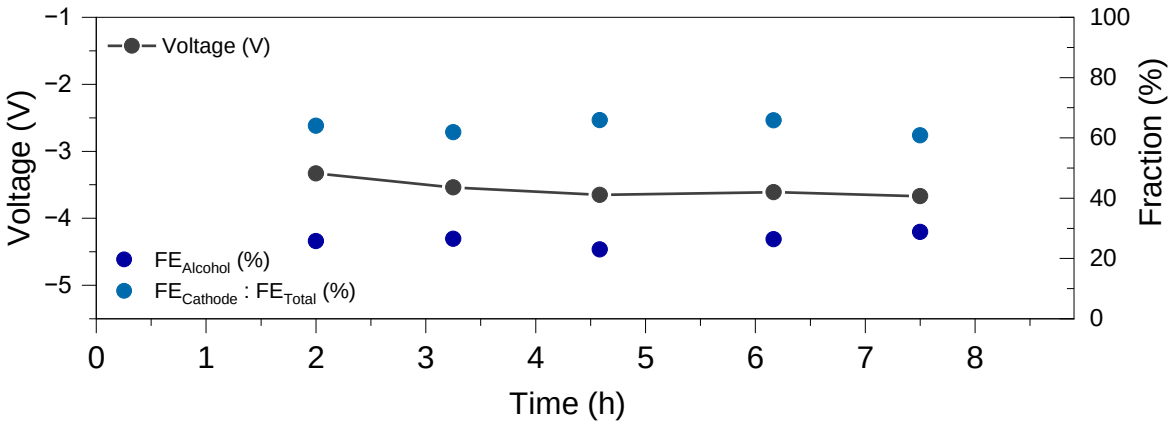
Supplementary Figure 13. Total product distribution for the 800 cm^2 cell test. The cell was operated at 100 mA/cm^2 with a Nafion 117 membrane. Source data for the results are provided as a Source Data file.



Supplementary Figure 14. The effect of feed gas pressure on ethanol crossover in the CEM MEA. Gas pressure was controlled using a back-pressure regulator attached to the outlet cathode gas. The electrolyzer were operated at 200 mA/cm^2. Source data for the results are provided as a Source Data file.

Supplementary Table 15. Time resolved faradaic efficiencies for each product during the long-term operation test with the 5 cm^2 electrolyzer, corresponding to Figure 4E.

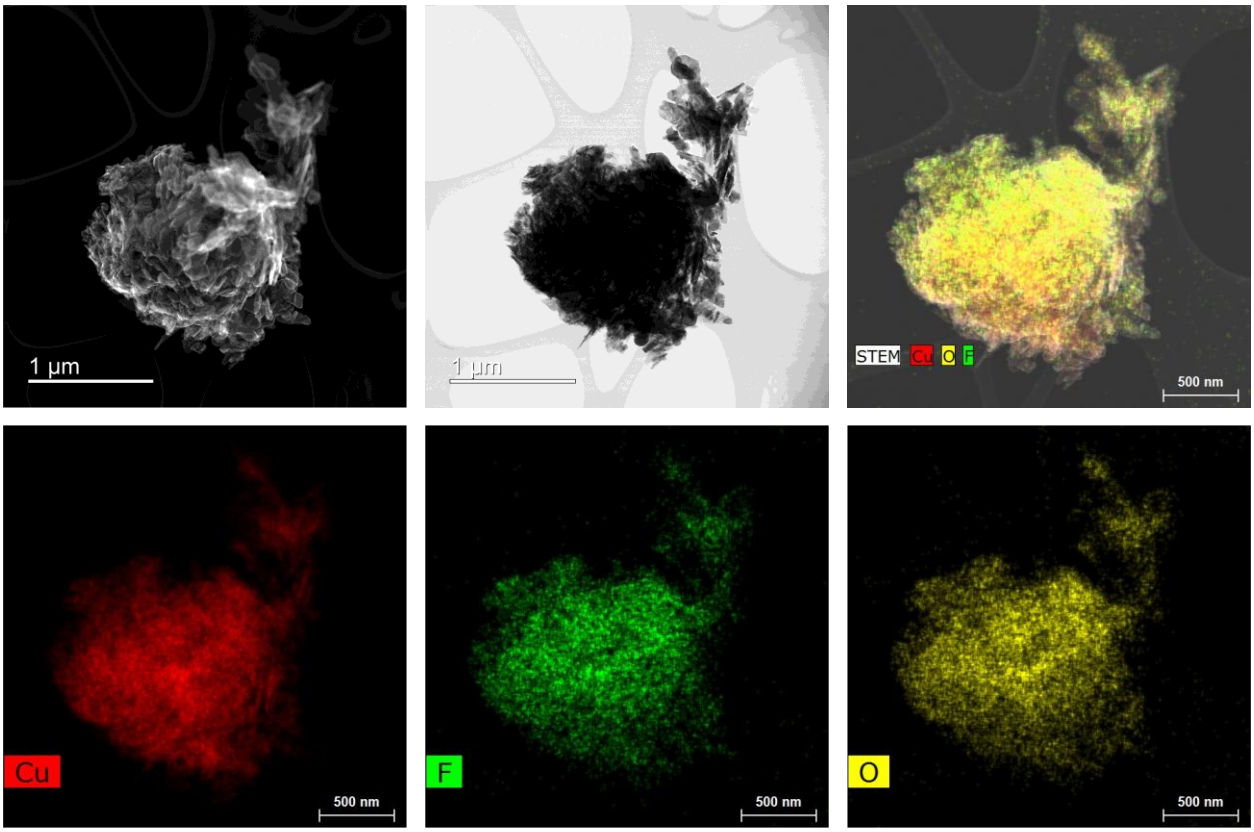
Time	H_2	CH_4	C_2H_4	CH_3COO^-	C_2H_5OH	C_3H_7OH
1 hour	14.4	0.07	25.8	15.8	20.4	20.1
25 hour	19.2	0.04	27.7	15.2	17.5	18.3
42 hour	19.1	0.04	26.8	15.6	20.1	17.2
48 hour	18.8	0.05	24.6	15.9	20.8	16.0
72 hour	21.4	0.07	24.4	17.5	19.4	16.8
80 hour	23.0	0.05	21.8	16.0	19.5	16.8
98 hour	21.5	0.06	22.8	18.9	18.0	15.8
101 hour	23.5	0.07	23.2	17.1	18.7	16.2



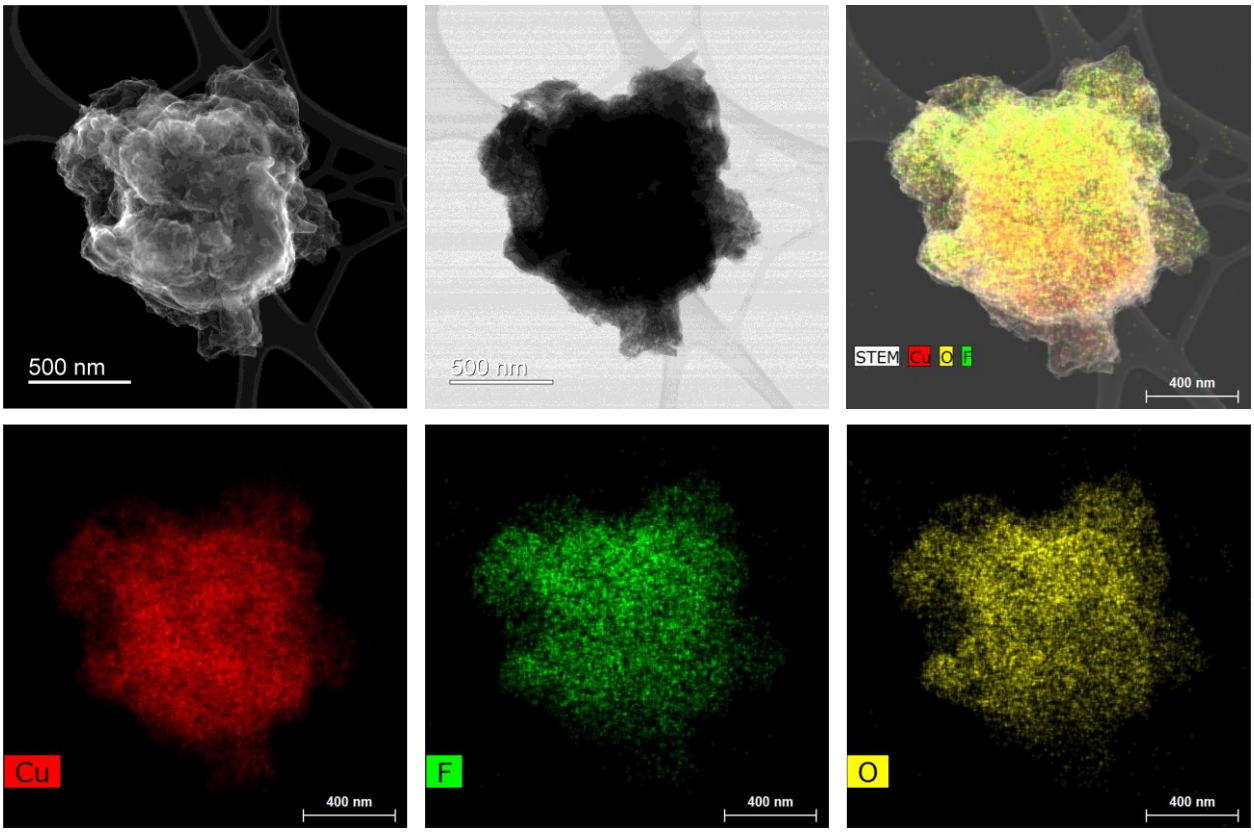
Supplementary Figure 15. Extended stability test of the 800 cm^2 electrolyzer. The electrolyzer was operated with a cold-trap for liquid product collection. The test was stopped after 7.5 hours due to lab restrictions on high CO flow rates for extended durations. Voltages represent the full-cell MEA potential and are not iR-corrected. Source data for the results are provided as a Source Data file.

Supplementary Table 16. Time resolved faradaic efficiencies for each product during the 7.5-hour test with the 800 cm^2 cell.

Time	H_2	CH_4	C_2H_4	CH_3COO^-	C_2H_5OH	C_3H_7OH
2h	39.7	0.10	16.9	12.8	21.5	4.2
3h15m	37.9	0.08	19.8	12.9	22.4	4.1
4h35m	41.8	0.1	21.9	7.5	19.3	3.7
6h10m	34.4	0.1	17.8	12.1	21.6	4.7
7h30m	40.8	0.09	16.4	9.5	22.5	6.2



Supplementary Figure 16. Scanning Transmission Electron Microscopy (STEM) images and the corresponding EDS mapping of the CuO catalyst with Nafion binder before the stability test. The fluorine distribution indicates the presence of Nafion binder, while the copper and oxygen distributions indicate CuO.



Supplementary Figure 17. STEM images and the corresponding EDS mapping of the CuO catalyst with Nafion binder after the 7.5-hour stability test with the 800 cm^2 cell. The fluorine distribution indicates the presence of Nafion binder, while the copper and oxygen distributions indicate CuO.

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